

University of Cape Town
Department of Mathematics and Applied Mathematics
Mathematics II Advanced Calculus (2AC)
Class Test 1 22nd March 2023

Time: 1.5 hrs

Full Mark: 37

Section A

Full Mark: 27

In this section full answers are expected. Marks will be deducted for incomplete solutions.

1. For each of the functions below say whether the limit exists at the point $(0, 0)$. If the limit exists, then find it.

(a) $f_1(x, y) := \frac{x^2 \sin x - y \sin y + y \sin x - x^2 \sin y}{y + x^2};$

(b) $f_2(x, y) := \frac{x \ln(1 + |y|)}{\sqrt{|x|^3 + y^2}};$

(c) $f_3(x, y) := \frac{y^2}{x^2} \sin\left(\frac{x}{\sqrt{x^2 + y^2}}\right).$

Hint: (b) $\ln(1 + t) \leq t$ for every $t \geq 0$.

[2, 3, 2]

2. Consider the vector function given by $\vec{r}(t) := \left(\frac{\ln(t^2 + 1)}{t^{\frac{4}{3}}}, e^{\frac{2t}{\sin t}}, \sqrt{1 + t} \arctan\left(\frac{1}{|t|}\right)\right).$
Find the limit $\lim_{t \rightarrow 0} \vec{r}(t).$

[2]

3. Let $\mathcal{C}$ be the curve described by the vector function $\vec{r}: \left[-\frac{\pi}{2}, \frac{3\pi}{4}\right] \rightarrow \mathbb{R}^3$ defined by

$$\vec{r}(t) := (2t^2 - 4, e^{-t} \sin(2t), e^{-t} \cos(2t)), \quad -\frac{\pi}{2} \leq t \leq \frac{3\pi}{4}.$$

- (a) Show that $\mathcal{C}$ is a regular curve.
(b) Show that $\vec{r}''(t) = (4, e^{-t}(-3 \sin(2t) - 4 \cos(2t)), e^{-t}(4 \sin(2t) - 3 \cos(2t))).$
(c) Find the curvature κ of the curve $\mathcal{C}$ at the point $(-4, 0, 1).$
(d) Find the unit tangent vector $\vec{T}$ at the point in (c).
(e) Write down the cartesian equation of the normal plane to the curve $\mathcal{C}$ at the point in (c).

[2, 1, 4, 1, 1]

4. Sketch the domain D of the function f defined by

$$f(x, y) := \sqrt{\ln\left(\frac{17}{1 + x^2 + y^2}\right)} - \ln(y^2 - x - 2).$$

Include all the steps explaining how you got your domain and the sketch! Make sure that the curves involved are clearly labelled and the domain found is also clearly shaded.

[4]

5. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $f(x, y) := \begin{cases} \frac{xy^2 - x^2}{\sqrt{x^2 + y^2}} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$

- (i) Study the continuity of the function f at the point $(0, 0)$.
- (ii) Say whether the partial derivatives $f_x(0, 0)$ and $f_y(0, 0)$ exist. If they exist then find them.
- (iii) Explain why the function f is not differentiable at $(0, 0)$.

[2, 2, 1]

Section B

Full Mark: 10

1. The curve $\mathcal{C}$ described by the vector function $\vec{r}(t) := \left(\frac{1 - (\ln t)^2}{1 + (\ln t)^2}, e^{2t}, \frac{2(\ln t)}{1 + (\ln t)^2} \right)$ with $t > 1$, lies
- (a) on the surface $\ln(\ln y) - xz = 0$;
 - (b) on the cylinder $x^2 + z^2 = 1$;
 - (c) in the set $\{x > 1\}$;
 - (d) on the sphere $x^2 + y^2 + z^2 = 1$;
 - (e) on the cone $y^2 = x^2 + z^2$.

[2]

2. The curve $\mathcal{C}$ described by the vector function $\vec{r}(t) := \left(2\sin^2(t), t, \frac{3}{2}\sin(2t) \right)$ with $-\pi \leq t \leq \frac{\pi}{4}$ has length $L(\mathcal{C})$ equal to

(a) $\int_{-\pi}^{\frac{\pi}{4}} \sqrt{4 + t^2 + 5\cos^2(2t)} dt$; (b) $\int_{-\pi}^{\frac{\pi}{4}} \sqrt{5\cos^2(2t)} dt$; (c) $\frac{5\pi\sqrt{5}}{4}$;

(d) $\int_{-\pi}^{\frac{\pi}{4}} \sqrt{4\sin^2(2t) + 1 + 9\cos^2(2t)} dt$; (e) $\int_{-\pi}^{\frac{\pi}{4}} \sqrt{4\sin^4(t) + t^2 + \frac{9}{4}\sin^2(2t)} dt$.

[2]

3. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $f(x, y) := \int_x^y \sin(t^2) dt$. Then its gradient, i.e., the vector having components its partial derivatives, at the point $\left(\frac{\sqrt{\pi}}{2}, \sqrt{\frac{3\pi}{2}} \right)$ is equal to:

(a) $\left(\frac{\sqrt{2}}{2}, -1 \right)$; (b) $\left(\frac{\sqrt{2}}{2}, 0 \right)$; (c) $\left(-\frac{\sqrt{2}}{2}, 1 \right)$; (d) $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right)$; (e) $\left(-\frac{\sqrt{2}}{2}, -1 \right)$.

[2]

4. The vector function $\vec{r}$ with expression $\vec{r}(t) := \left(\ln(4 - t^2), \frac{1}{\sqrt{2 - e^t}}, \frac{3t}{2t^2 - 4t + 3} \right)$ has domain $\text{Dom}(\vec{r})$ equal to the interval:

(a) $(-2, \ln 2)$; (b) $(\ln 2, 2)$; (c) $(\ln 2, e^4)$; (d) $(-4, \ln 2)$; (e) $[-2, \ln 2]$.

[2]

5. The level curve of the function $f(x, y) = \sin(xy) - \cos(xy)$ passing through the point $(-1, \pi)$

- (a) does intersect the y -axis; (b) also passes through the point $(-\sqrt{\pi}, \frac{\sqrt{\pi}}{2})$;
(c) has cartesian equation $\cos(xy) - \sin(xy) = -1$; (d) also passes through the point $(\frac{\sqrt{\pi}}{2}, \frac{\sqrt{\pi}}{2})$;
(e) intersects the x -axis at the point $(-\frac{\pi}{2}, 1)$.

[2]